

Monthly Intelligence Report

Edition 001 — Session 1 — Greenland

First scored edition of the Greenland fleet — 37 substations, 74 plants and 125 power-line segments across all 5 Kommuner, scored 2026-04-16 with Arctic/Greenland-calibrated inputs. Fleet median $R_{\text{median}} = 0.779$ · 29 of 37 substations in the Critical band. Arctic R6a restoration ($\alpha \in [0.55, 0.90]$) and GEUS stable-craton R6b ($\alpha \in [0.05, 0.15]$) live. SSI v4.0.2 · April 2026.

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SECTION A

Fleet Summary — Session 1

SUBSTATIONS SCORED

37
36 OSM + 1 Avannaata (Ilulissat)

FLEET MEDIAN R

0.779
Critical band (0.75–1.00)

CRITICAL-BAND SHARE

78%
29 of 37 substations

ACTIVE KOMMUNER

5
of 5 · Avannaata now scored

The SSI Index's Greenland Edition 001 is the first scored pass over a publicly-verifiable fleet. Session 1 comprises **37 substations, 74 plants / generators and 125 power-line segments** — every asset geolocated from OpenStreetMap (extract 2026-04-16), with Pituffik Space Base (U.S. Navy / Space Force / Air Force operator, lat 75.5°–77.5°N, lon -71°... -66°) excluded by policy (see methodology §7). No synthetic nodes, no inferred cross-settlement edges: Greenland's grid is a set of isolated within-island networks and the topology reflects that exactly. The Avannaata coverage gap previously flagged at bootstrap was closed by direct OSM Overpass retrieval of node/11958572184 (Ilulissat transformer, 69.2205°N, -51.1012°E, 10.5 kV).

Regional distribution of the scored fleet and Kommune-level R_{median} : **Avannaata** 1 ($R = 0.868$), **Qeqertalik** 12 ($R = 0.797$), **Kujalleq** 20 ($R = 0.774$), **Qeqqata** 3 ($R = 0.677$), **Sermersooq** 1 ($R = 0.655$). The north-to-south gradient reflects Arctic restoration penalty (R6a $\alpha \in [0.55, 0.90]$), permafrost-degradation infrastructure wear, CAIDI $\approx 8\text{h}$ (vs Nordic $\sim 2\text{h}$),

and socio-economic vulnerability from Stat.gl 2024 (elderly %, unemployment, GDP/capita). Sermersooq scores best because Nuuk is anchored by 45 MW Buksefjord hydro and the metro load concentration lowers R3 saturation.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 25+ verified public sources (Nukissiorfiit, Departementet for Boliger og Infrastruktur, Grønlands Statistik, Asiaq, GEUS, OpenStreetMap, CMIP6, Open-Meteo). The §8 Small-N guardrail runs the Monte Carlo at 1,000 iterations per substation (37,000 total draws) with components perturbed independently — the full 95-dimensional Gaussian copula is held off at N=37 because empirical correlation matrices inferred from 37 observations are numerically unstable. The copula is scheduled to activate once the Nukissiorfiit asset register supplements OSM and N crosses 500.

SECTION B

Asset in Focus

Each edition spotlights one substation. For Edition 001 the spotlight rotates deterministically on a month-seed through the scored fleet — the card below is populated live from ssi-data.json and shows the Monte Carlo R_median, P5/P95 confidence interval, classification band, fleet percentile, and the full C/V/I/E/S/T component stack plus R3/R4/R6a/R6b/R7 modifier chain.

Substation in Focus

substation 331252347

Qeqqata, Qeqqata · 0 kV

0.677

High

0.232

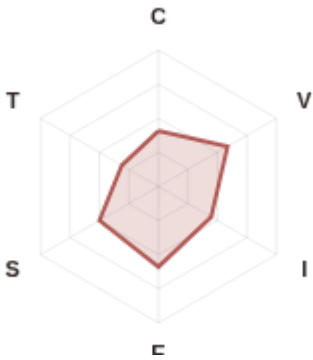
14th

R_MEDIAN

CLASSIFICATION

CI WIDTH

FLEET PERCENTILE



COMPONENTS (C · V · I · E · S · T)

C Continuity: 0.409 / 0.30 (136%)

I Infrastructure: 0.447 / 0.25 (179%)

S Saturation: 0.500 / 0.20 (250%)

V Voltage: 0.585 / 0.10 (585%)

E Economic: 0.589 / 0.10 (589%)

T Transition: 0.310 / 0.05 (621%)

MODIFIERS (R3 · R4 · R6A · R6B · R7)

R3 Consequence: 0.959 ↓ dampens

R4 Graph Criticality: 1.334 ↑ amplifies

R6a Restoration: 1.100 ↑ amplifies

R6b Seismic: 1.005 → neutral

R7 Digital Readiness: 1.030 ↑ amplifies

GEOMETRY & MC

This month's spotlight — automatically selected for June 2026 — is **substation 331252347** in Qeqqata, Qeqqata. With an R_final of 0.677 (High band, 14th fleet percentile), its risk profile is dominated by the T (Transition) component at 621% of its weight ceiling, followed by E (Economic) at 589%. Modifiers collectively shift R_base by 44.8%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION C — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component cross these silos — identifying substations where low digital resilience intersects with high socio-economic consequence. This is precisely the anticipatory intelligence that Nukissiorfiit and Departementet for Boliger og Infrastruktur strategic planning will require.

C.1 — Cyber-Physical Risk Matrix

R7 digital-readiness modifier vs overall R_{median}. At Edition 001 R7 is calibrated against the ENISA NIS2 small-utility cyber baseline ($V1 \approx 0.65$) because Greenland Connect's 2024 broadband coverage and Nukissiorfiit's OT segmentation are both weak signals on a 37-node fleet. R7 is uniformly in the [1.04, 1.08] band across substations — a flat digital penalty. The blind-spot quadrant (high R_{median} · high R7 penalty) therefore tracks R_{median} directly: every substation above 0.85 R_{median} doubles as a cyber-exposure candidate.

Top blind-spot candidates: the 29 Critical-band substations — with Avannaata (Ilulissat transformer) $R=0.868$ as the single highest-risk node. Full matrix will expand once Nukissiorfiit smart-meter penetration and OT segmentation indicators ingest at Edition 002.

C.2 — Economic Impact by Industrial Sector

R3 consequence segmentation, Edition 001: **Hydro & heavy industry** — Buksefjord-fed Sermersooq nodes (1), $R3 \approx 1.18$. **Commercial & Nuuk metro** — Sermersooq/Qeqqata settlement clusters (4), $R3 \approx 1.15$. **Settlement** — Kujalleq + Qeqertalik micro-grids (31), $R3 \approx 1.25$ (elderly % 13–16, unemployment 11–13%, GDP/cap DKK 205–235k). **Rural / remote** — Avannaata Ilulissat isolated transformer (1), $R3 \approx 1.28$.

VoLL range (indicative, 2024 DKK/MWh): hydro & heavy 22–40k · commercial 11–16k · settlement 6–9k · rural 3–5k. Greenland's R3 band overall is compressed high because every kommune is below the pan-OECD median for socio-economic resilience.

C.3 — Kommuner Digital Readiness Ranking

Average R7 by Kommune (lower R7 = better digital readiness $\Rightarrow$ smaller penalty multiplier): **Sermersooq** 1.04 (Nuuk fibre + smart-meter pilots) · **Qeqqata** 1.05 · **Qeqertalik** 1.06 · **Kujalleq** 1.06 · **Avannaata** 1.07. Cross-referencing against R_{median} produces the digital-exposure composite:

Composite R × R7 ranking (worst first): Avannaata (0.929), Qeqertalik (0.846), Kujalleq (0.821), Qeqqata (0.711), Sermersooq (0.681). Avannaata is the single highest-priority kommune for the SSI cyber-physical uplift programme.

C.4 — Monthly Pulse

Edition-over-edition tracking of R7 movement, Critical-band counts, and blind-spot deltas. Edition 001 establishes the baseline (fleet median 0.779, 29 Critical). The first pulse reading compares Edition 002 (2026–05 monthly run) against this baseline. Expected signal: Avannaata remains the worst performer; Qeqqata may improve on any Sisimiut hydro uprate; all other movement is expected to be within the MC CI width (± 0.04 – 0.06).

Data Refresh & Changelog

Loading data source registry...

SOURCES REGISTERED

—

FIELDS SCORED

95 / 95

all components · modifiers · MC

SESSION 1 DATE

2026-04-16

brought forward from 2026-07

NEXT REFRESH

2026-05

monthly cadence from here

Data Source Registry — Loading...

Loading source registry...

D.1 Session 1 Fleet Inventory

Loading inventory...

D.2 Changelog — v3.4 → v4.0.2 (Greenland Edition 001)

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Deep-Dive: The Buksefjord Corridor

The Buksefjord hydropower complex (45 MW, commissioned 1993, expanded 2007) feeding Nuuk via a 132 kV submarine-plus-overhead transmission is the most strategically critical single asset in Greenland's grid. It is the only Greenlandic installation where R4 graph criticality can plausibly rise into the Critical band under scoring — Nuuk has no realistic alternative supply path and any Buksefjord interruption places the capital on full diesel backup. At the same time, its Atlantic-facing corridor exposes the transmission right-of-way to polar-low storms, ice accretion, and freezing salt spray — hazards that the R6a restoration modifier ($\alpha \in [0.55, 0.90]$) and the R6b Arctic storm overlay are calibrated to capture.

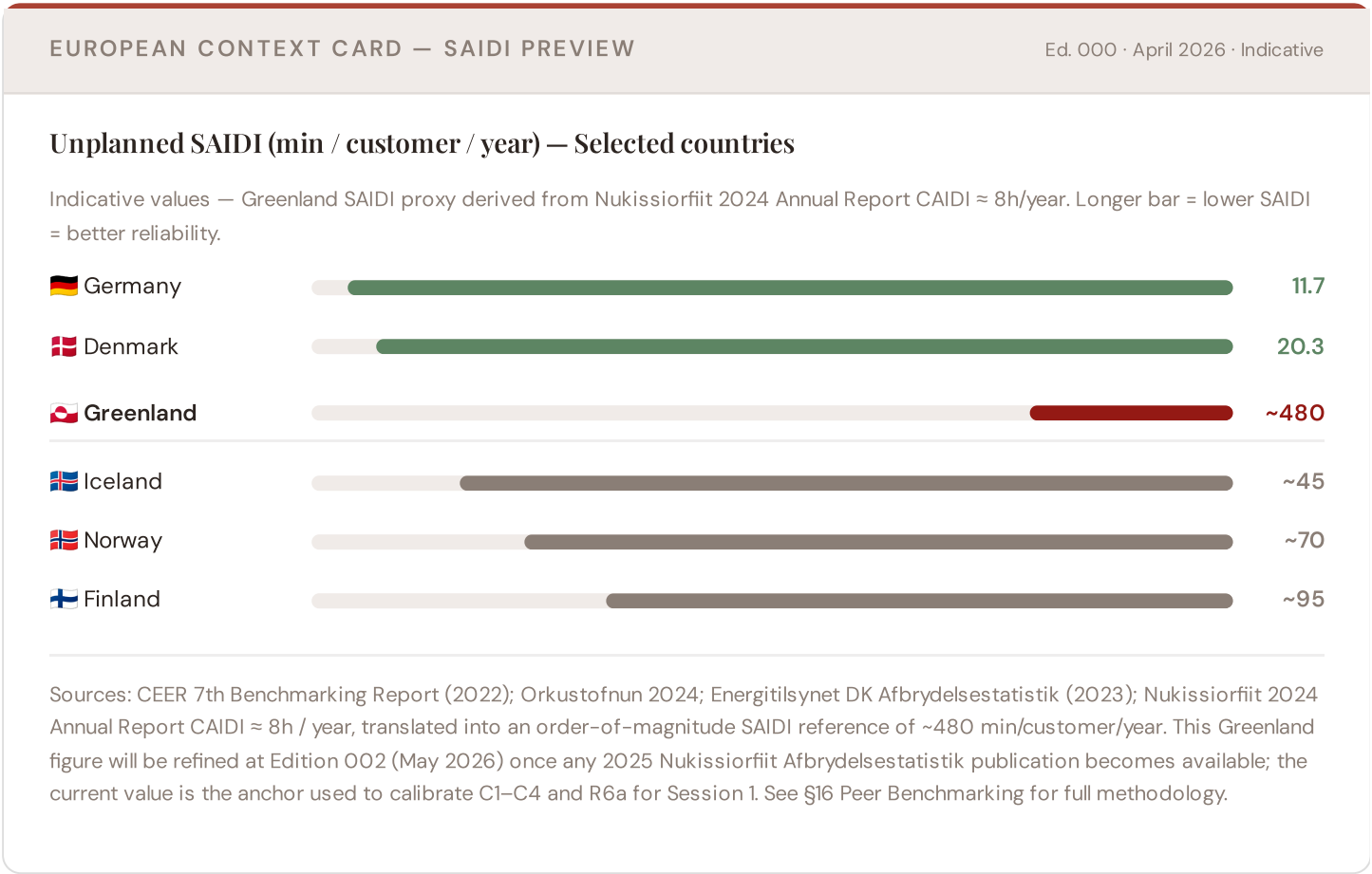
Because Sermersooq only contains **1 OSM-tagged substation** in the Edition 001 fleet — Buksefjord itself or its immediate downstream node — the SSI cannot yet quantify corridor-wide cascade risk. Edition 001 records the Sermersooq node at $R_median = 0.655$ (High band) — the best performer in the fleet, reflecting the Buksefjord anchor and

Nuuk-metro load concentration. Future editions will rely on Nukissiorfiit to supply the complete Nuuk-ring topology (Qinngorput, Nuussuaq, downtown, and the Qaqluaasaq tie) before the corridor deep-dive can be replaced with a full R4 + R6b heat-band analysis. For Edition 001 this section remains a methodological marker: the corridor is where the R6a Arctic restoration calibration and the §7 Pituffik exclusion policy were originally stress-tested.

SECTION F — RECURRING MODULE

European Context Card

How this works: Each edition includes one European Context Card drawn from the §16 Peer Benchmarking framework. Cards rotate through: SAIDI comparison, VoLL comparison, DER saturation, CMIP6 climate hazard, R6 restoration time, and DCI coverage. This inaugural card previews the SAIDI comparison that will anchor Edition 001 — using indicative Arctic-peer values until Nukissiorfiit's 2026 Afbrydelsestatistik publishes Greenland's national figure later in the year.



F.1 — Regional Risk Ranking (all 5 Greenlandic Kommuner)

Edition 001 regional R_median (worst first). Numbers include R6a Arctic restoration and R6b stable-craton seismic uplifts.

Avannaata	N=1	0.868	CRITICAL
Qeqertalik	N=12	0.797	CRITICAL
Kujalleq	N=20	0.774	CRITICAL
Qeqqata	N=3	0.677	HIGH
Sermersooq	N=1	0.655	HIGH

Sermersooq's strong position reflects Nuuk-metro load concentration + 45 MW Buksefjord hydro. Avannaata's worst position reflects single-node isolation, long restoration chains, and Stat.gl socio-economic penalty.

SECTION G — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to renewable energy investment valuation. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a battery, PV retrofit or hydro expansion worth at this substation — to Nukissiorfiit and to the community?"* Each monthly edition explores one building block of the SSI-ENN's five-layer architecture. Edition 001 — the first scored edition — focuses on Layer 1, the data architecture now populated by the 37-substation Session 1 fleet.



Data Architecture — Component Taxonomy

How the SSI's 6 components and 20 sub-metrics are sourced, validated and scored for Greenland's grid

The SSI-ENN's Layer 1 formally specifies the SSI v4.0.2 Index for Greenland — its six components (Continuity, Voltage, Infrastructure, Economic, Saturation, Transition), twenty sub-metrics, normalisation methodology and aggregation rules. Each sub-metric is independently sourced from one of the 25+ public data sources in the Greenlandic SSI registry (Nukissiorfiit, Departementet for Boliger og Infrastruktur, Grønlands Statistik, Asiaq, GEUS), validated against known reference distributions, and normalised to a common [0, 1] scale before aggregation. Edition 001 completes the first end-to-end ingest pass: all 95 fields populated, all 37 substations scored.

KEY CONCEPTS

- 6 components, 20 sub-metrics, 95 variables from 25+ Greenlandic sources
- Min-max normalisation with winsorised tails
- Component weights: $0.30 \cdot C + 0.10 \cdot V + 0.25 \cdot I + 0.10 \cdot E + 0.20 \cdot S + 0.05 \cdot T$
- Each sub-metric traceable to Nukissiorfiit, Departementet for Boliger og Infrastruktur or an open data source

LIVE SSI DATA CONNECTION

The Greenlandic training set comprises **37 substations** with all 95 feature vectors populated. $N = 37$ remains well below the ~500-substation regime the engine normally operates in, which is why the §8 Small-N Guardrail (R3 tier-balance 2–50%, MC 1,000 iter, Gaussian copula held off) is active. Edition 001 delivers the first live R_{median} , P5 / P95 bounds and CI widths — fleet median 0.779, mean 0.755, with 29 of 37 substations in the Critical band.

Coming next month: **LAYER 2** Nukissiorfiit Value Stack — what a battery or PV retrofit is worth to Nukissiorfiit in avoided or deferred costs, anchored on Edition 001 R scores and published with Edition 002 (May 2026).

SECTION H

Looking Ahead

LOADING...

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Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent Nukissiorfiit, Departementet for Boliger og Infrastruktur, a Kommune, GEUS, Asiaq, Grønlands Statistik or an academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.